## DATA 3464: Fundamentals of Data Processing

# Categorical Data Basics

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# Topic overview

- Exploring categorical data
- Basic categorical data encoding strategies
- Out of scope: Fancier encoding strategies, like feature hashing and target encoding

**Resources used:**

- Feature Engineering Chapter 5
- Scikit-learn User Guide (7.3)

# What is categorical data?

- Samples can take on one of several discrete values or groups
  - **Nominal**: no particular order to the groups
  - **Ordinal**: groups relate to each other in a specific order
- Categories can be represented as strings *or* numeric types
  - Domain knowledge is necessary!

> *Let's take a few minutes to brainstorm some examples*

# Exploring categorical data

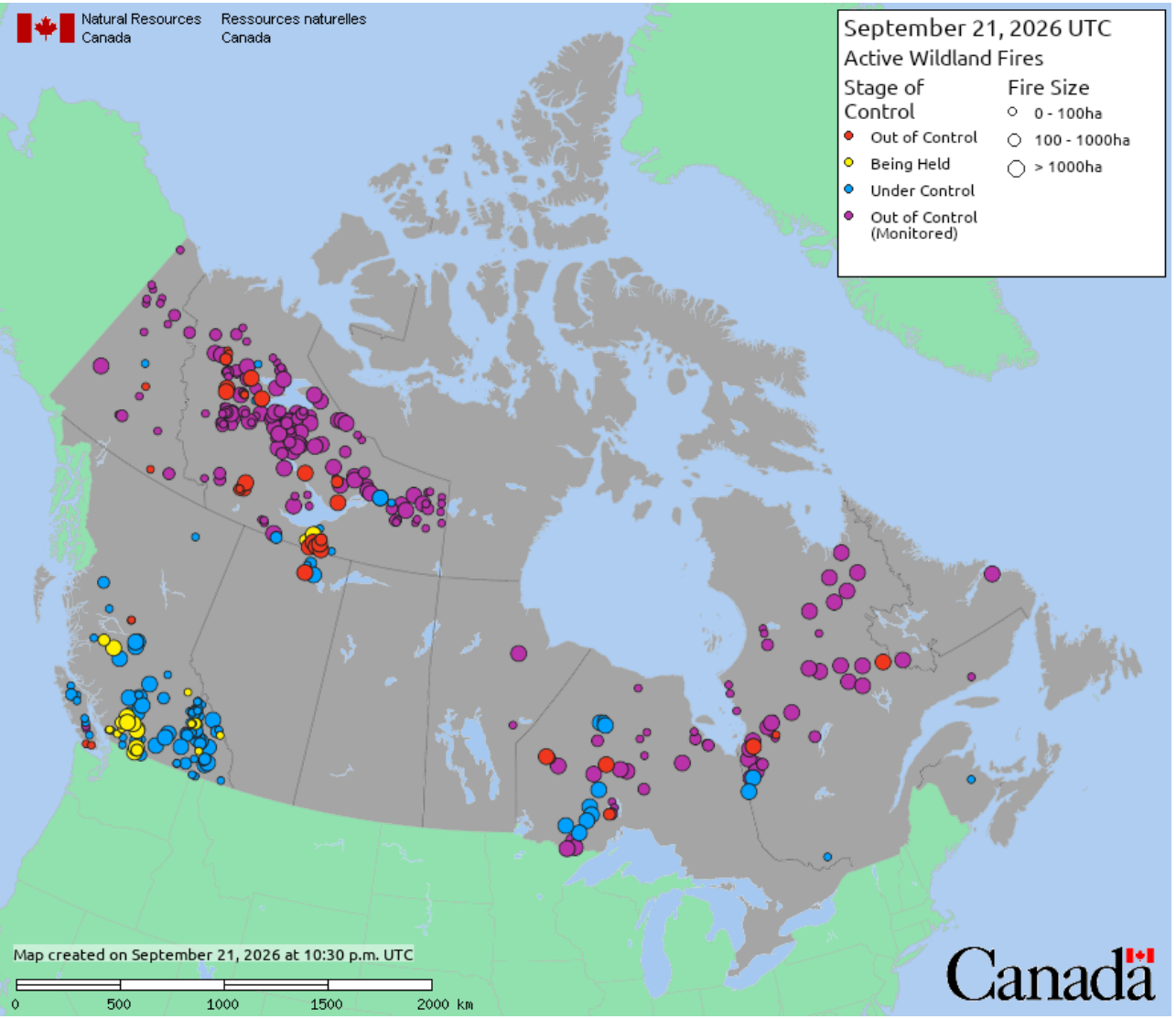
We've already done some of this, but some ideas to consider:

- `pandas.DataFrame.value_counts()` - how many of each category?
- Use category to group, then compute summary stats
- Plot color per category
- Scatter plot with jitter

# Representing categorical data

- To do math on categorical data, we need to map to numbers
- Consider:
  - Ordinal or nominal?
  - How many possible categories (cardinality)?
  - Any chance new ones might show up?

> *How could we encode the examples?*
>
> *What are the benefits and drawbacks of each method?*




# Ordinal encoding

| Category | Feature |
|---|---|
| Under Control | 0 |
| Being Held | 1 |
| *Out of Control (Monitored) | 2 |
| Out of Control | 3 |

* I am not a wildfire expert, this is my best guess at ordering

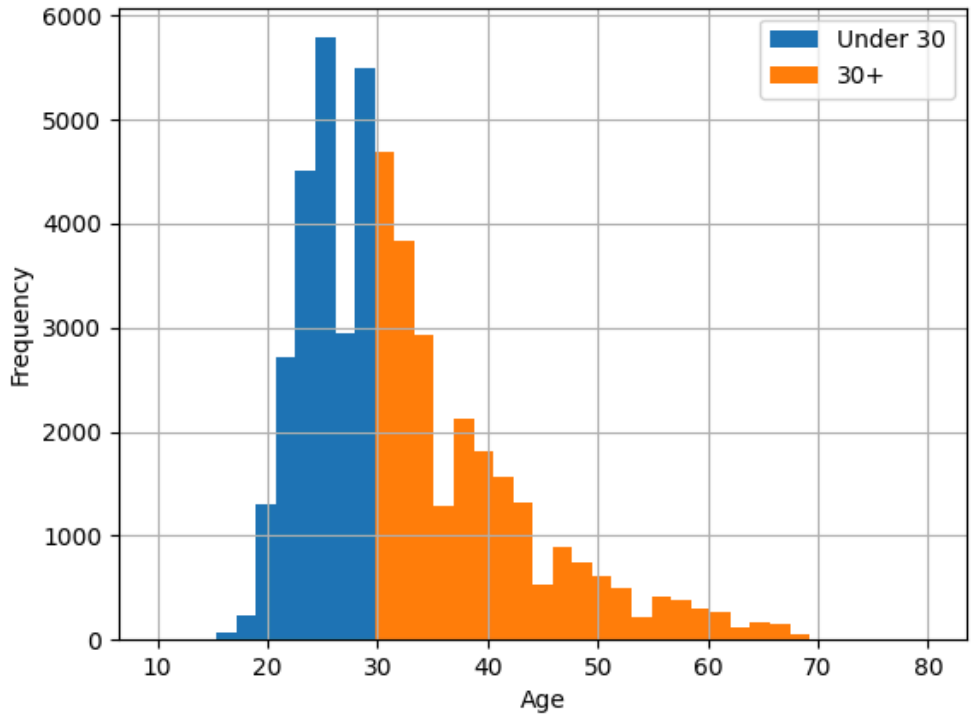
Image source: Natural Resources Canada

# Nominal categories: one-hot encoding

- Categories have no natural relationship
- Create $k$ new features from $k$ categories, very sparse matrix

| **Animal** | | **cat** | **dog** | **rabbit** |
|---|---|---|---|---|
| cat | → | 1 | 0 | 0 |
| dog | → | 0 | 1 | 0 |
| rabbit | → | 0 | 0 | 1 |

# From numbers to categories: discretization




- Sometimes numeric data is better represented as categorical
- Still needs encoding strategy
- Can introduce nonlinear relationships
- Like everything, contextual

# Advantages/disadvantages of various methods

## Ordinal

+ Compact (1 column/feature)
+ Only useful for naturally ordered data
- Imposes (linear) relationship
- Difficult to deal with novel categories

## One-hot

+ No assumptions made about relationships
+ Novel features implied by 0
- Memory intensive
- Not good for high cardinality
- Potential collinearity issues

# Coming up next

- Lecture: Basic numeric transformations
- Lab: Modelling with categorical data
- Assignment 1 due soon!

> *Feature Engineering Chapter 5*